SAT Math Mastery · Session 4

Equivalent Expressions and Nonlinear Equations

Time 3 hours: Part A in class (about 2 hours 20 minutes of solving), then Part B, the self-assessment, done independently (about 45 minutes), and a short error log.

Topics quadratic equations: discriminant, sum and product of solutions · absolute-value and rational equations · equivalent rational expressions · exponent rules and radicals · factoring and polynomial division · matching coefficients · undefined values

Name / Date _____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos, one expression per line; tap a grey dot to read an intersection, intercept or vertex. A letter such as a, c or k becomes a slider when you accept Desmos's offer to add one.

1. HARD · NO CALCULATOR NEEDED

For nonzero real numbers p and q, the equation $px^2 + 3qx + 4p = 0$ has exactly one real solution, and $pq = -48$. If $p > 0$, what is the value of $p + q$?

(A) -14 **(B) -2 ✓**

(C) 2 (D) 14

1. Exactly one real solution means the discriminant is zero: $(3q)^2 - 4(p)(4p) = 0$, which is $9q^2 = 16p^2$. So $3q = \pm 4p$, that is $q = \pm\frac{4p}{3}$.

2. The product $pq = -48$ is negative and $p > 0$, so $q < 0$. Take $q = -\frac{4p}{3}$: then $pq = -\frac{4p^2}{3} = -48$, so $p^2 = 36$.

3. Since $p > 0$, $p = 6$, and $q = -\frac{4 \cdot 6}{3} = -8$.

4. $p + q = 6 + (-8) = -2$. Option (B).

Answer: (B)

2. HARD · NO CALCULATOR NEEDED

The equation $x^2 - 9x + k = 0$, where k is a constant, has two real solutions that differ by 3. What is the value of k?

1. Call the solutions r and $r + 3$. Their sum is the opposite of the x-coefficient, so it is 9.

2. $r + (r + 3) = 9$ gives $2r = 6$, so $r = 3$. The solutions are 3 and 6.

3. The constant term is the product of the solutions: $k = 3 \cdot 6 = 18$.

4. Check: $(x - 3)(x - 6) = x^2 - 9x + 18$.

Answer: 18

3. HARD · NO CALCULATOR NEEDED

The equation $2x^2 - 10x + c = 0$, where c is a constant, has a product of solutions equal to 6. What is the sum of the reciprocals of the two solutions?

(A) $\frac{1}{5}$ (B) $\frac{1}{30}$

(C) $\frac{5}{6}$ ✓ (D) $\frac{6}{5}$

1. For $ax^2 + bx + c = 0$ the solutions have sum $-\frac{b}{a}$ and product $\frac{c}{a}$. Here the sum is $\frac{10}{2} = 5$.

2. The product is $\frac{c}{2} = 6$, so $c = 12$. The solutions are real because the discriminant is $10^2 - 4 \cdot 2 \cdot 12 = 4 > 0$.

3. Add the reciprocals over a common denominator: $\frac{1}{r} + \frac{1}{s} = \frac{r+s}{rs}$.

4. $\frac{r+s}{rs} = \frac{5}{6} = \frac{5}{6}$. Option (C).

Answer: (C)

4. HARD · NO CALCULATOR NEEDED

Let r and s be the two solutions of $x^2 - 7x + 3 = 0$. What is the value of $r^2 + s^2$?

1. For $x^2 - 7x + 3 = 0$ the sum of the solutions is $r + s = 7$ and the product is $rs = 3$.

2. Square the sum: $(r + s)^2 = r^2 + 2rs + s^2$, so $r^2 + s^2 = (r + s)^2 - 2rs$.

3. $r^2 + s^2 = 7^2 - 2 \cdot 3 = 49 - 6 = 43$.

Answer: 43

5. HARD · NO CALCULATOR NEEDED

The equation $x^2 - 5x + 6 = 0$ has solutions p and q. Which of the following equations has solutions $2p$ and $2q$?

(A) $x^2 - 10x + 24 = 0$ ✓ (B) $x^2 - 10x + 12 = 0$

(C) $x^2 - 5x + 12 = 0$ (D) $2x^2 - 5x + 6 = 0$

1. Factor: $x^2 - 5x + 6 = (x - 2)(x - 3)$, so $p = 2$ and $q = 3$.

2. The new solutions are $2p = 4$ and $2q = 6$. Their sum is 10 and their product is 24.

3. An equation with sum S and product P is $x^2 - Sx + P = 0$, so it is $x^2 - 10x + 24 = 0$.

4. Check: $(x - 4)(x - 6) = x^2 - 10x + 24$. Option (A).

Answer: (A)

6. HARD · NO CALCULATOR NEEDED

The equation $x^2 - bx + 27 = 0$, where b is a positive constant, has two real solutions, and one solution is 3 times the other. What is the value of b?

1. Call the solutions r and $3r$. The product of the solutions is the constant term, and the sum is b.
2. Product: $r \cdot 3r = 3r^2 = 27$, so $r^2 = 9$ and $r = \pm 3$.
3. Sum: $b = r + 3r = 4r$. Since b is positive, take $r = 3$: $b = 12$.
4. Check: $x^2 - 12x + 27 = (x - 3)(x - 9)$.

Answer: 12

7. HARD · DESMOS

$$x^2 - 6x = 2|x - 3|$$

What is the sum of all real solutions of the equation above?

(A) 9

(B) 0

(C) 3

(D) 6 ✓

1. Type `y=x^2-6x` and `y=2abs(x-3)`.
2. Tap both intersections: $(-1.1623, 8.3246)$ and $(7.1623, 8.3246)$.
3. Type `-1.1623+7.1623`: it returns 6, so the sum of the solutions is 6.
4. Why: the parabola and the V both have their axis at $x = 3$, so the two points are the same distance from 3 and their x-values add to $2 \cdot 3$. Option (D).

Answer: (D)

8. HARD · DESMOS

$$\frac{x}{x-4} = 2 + \frac{4}{x-4}$$

How many solutions does the equation above have?

(A) Zero ✓

(B) Exactly one

(C) Exactly two

(D) Infinitely many

1. Type `y=x/(x-4)` and `y=2+4/(x-4)`.
2. The curves never touch. The second is always above the first: type `2+4/(5-4)`: it returns 6 and type `5/(5-4)`: it returns 5 at $x = 5$, and type `2+4/(8-4)`: it returns 3 and type `8/(8-4)`: it returns 2 at $x = 8$.
3. Algebra agrees. Multiply by $x - 4$: $x = 2(x - 4) + 4$, so $x = 2x - 4$ and $x = 4$.
4. But $x = 4$ makes the denominator 0, so it is not allowed. The equation has no solution. Option (A).

Answer: (A)

9. HARD · NO CALCULATOR NEEDED

The function f is defined by $f(x) = ax^2 + bx + c$, where a, b, and c are constants. The graph of $y = f(x)$ in the xy-plane passes through $(1, 8)$ and $(-1, 2)$ and has exactly one x-intercept. What is the greatest possible value of a?

(A) $\frac{5}{2}$

(B) 9

(C) $\frac{9}{2}$ ✓

(D) $\frac{1}{2}$

1. $f(1) = a + b + c = 8$ and $f(-1) = a - b + c = 2$. Subtract: $2b = 6$, so $b = 3$. Then $a + c = 5$, so $c = 5 - a$.

2. One x-intercept means the discriminant is zero: $b^2 - 4ac = 0$, so $9 - 4a(5 - a) = 0$, which is $4a^2 - 20a + 9 = 0$.

3. Factor: $(2a - 1)(2a - 9) = 4a^2 - 20a + 9$, so $a = \frac{1}{2}$ or $a = \frac{9}{2}$.

4. The greatest possible value is $a = \frac{9}{2}$. Option (C).

Answer: (C)

10. HARD · NO CALCULATOR NEEDED

The expression $\frac{3}{x-4} - \frac{2}{x+1}$ is equivalent to which of the following, for $x \neq 4$ and $x \neq -1$?

(A) $\frac{x+11}{x^2+3x-4}$

(B) $\frac{x+11}{x^2-3x-4}$ ✓

(C) $\frac{x-11}{x^2-3x-4}$

(D) $\frac{5x-5}{x^2-3x-4}$

1. Use the common denominator $(x - 4)(x + 1) = x^2 - 3x - 4$.

2. Rewrite each fraction: $\frac{3(x+1)}{(x-4)(x+1)} - \frac{2(x-4)}{(x-4)(x+1)}$.

3. The numerator is $3(x + 1) - 2(x - 4) = 3x + 3 - 2x + 8 = x + 11$. Watch the sign on $-2 \cdot (-4)$.

4. Check at $x = 0$: $\frac{3}{-4} - \frac{2}{1} = -\frac{11}{4}$ and $\frac{11}{-4} = -\frac{11}{4}$. Option (B).

Answer: (B)

11. HARD · NO CALCULATOR NEEDED

The expression $\frac{5}{x+2} + \frac{3x}{x^2+x-2}$ is equivalent to which of the following, for $x \neq -2$ and $x \neq 1$?

(A) $\frac{8x-5}{x^2-x-2}$

(B) $\frac{3x+5}{x^2+x-2}$

(C) $\frac{8x+5}{x^2+x-2}$

(D) $\frac{8x-5}{x^2+x-2}$ ✓

1. Factor the second denominator: $x^2 + x - 2 = (x + 2)(x - 1)$. This is the common denominator.

2. Only the first fraction needs rewriting: $\frac{5}{x+2} = \frac{5(x-1)}{(x+2)(x-1)} = \frac{5x-5}{x^2+x-2}$.

3. Add the numerators: $(5x - 5) + 3x = 8x - 5$. So the sum is $\frac{8x-5}{x^2+x-2}$.

4. Check at $x = 0$: $\frac{5}{2} + 0 = \frac{5}{2}$ and $\frac{-5}{-2} = \frac{5}{2}$. Option (D).

Answer: (D)

12. HARD · NO CALCULATOR NEEDED

The operation $\odot$ is defined for all real numbers $x \neq 1$ by $x^{\odot} = \dfrac{x+3}{x-1}$. Which of the following is equivalent to $x^{\odot} - 1$?

(A) $\dfrac{4}{x-1}$ ✓

(B) $\dfrac{2x+4}{x-1}$

(C) $\dfrac{2}{x-1}$

(D) $\dfrac{x+2}{x-1}$

1. Write 1 with the same denominator: $1 = \dfrac{x-1}{x-1}$.

2. Subtract the numerators and keep the brackets: $(x+3) - (x-1) = x + 3 - x + 1 = 4$.

3. So $x^{\odot} - 1 = \dfrac{4}{x-1}$.

4. Check at $x = 3$: $\dfrac{6}{2} - 1 = 2$ and $\dfrac{4}{2} = 2$. Option (A).

Answer: (A)

13. HARD · NO CALCULATOR NEEDED

If $a(x-2)(x+6) = a(x-h)^2 + k$ for all values of x, where $a \neq 0$, what is k in terms of a?

(A) $16a$

(B) $-16a$ ✓

(C) $-12a$

(D) $4a$

1. The expression $a(x-2)(x+6)$ has zeros 2 and -6; the vertex form $a(x-h)^2 + k$ has its vertex halfway between them.

2. $h = \dfrac{2 + (-6)}{2} = -2$.

3. k is the value at the vertex: $a(-2-2)(-2+6) = a(-4)(4) = -16a$.

4. Check: $a(x+2)^2 - 16a = a(x^2 + 4x + 4) - 16a = a(x^2 + 4x - 12) = a(x-2)(x+6)$. Option (B).

Answer: (B)

14. HARD · NO CALCULATOR NEEDED

Which expression is equivalent to $x^4 - 2x^2 - 15$?

(A) $\left(x^2 - 15\right)\left(x^2 + 1\right)$

(B) $\left(x^2 - 1\right)\left(x^2 + 15\right)$

(C) $\left(x^2 - 5\right)\left(x^2 + 3\right)$ ✓

(D) $\left(x^2 + 5\right)\left(x^2 - 3\right)$

1. Let $u = x^2$. The expression becomes $u^2 - 2u - 15$.

2. Find two numbers with product -15 and sum -2: they are -5 and 3. So $u^2 - 2u - 15 = (u-5)(u+3)$.

3. Put x^2 back: $(x^2 - 5)(x^2 + 3)$.

4. Check the middle term: $x^4 + 3x^2 - 5x^2 - 15 = x^4 + (-2)x^2 - 15$. Option (C).

Answer: (C)

15. HARD · NO CALCULATOR NEEDED

If $\dfrac{x^{a^2}}{x^{b^2}} = x^{20}$, $x > 1$, and $a + b = 4$, what is the value of $a - b$?

1. For $x > 1$, equal powers have equal exponents. Dividing powers subtracts exponents: $a^2 - b^2 = 20$.

2. Factor the difference of squares: $(a - b)(a + b) = 20$.

3. Substitute $a + b = 4$: $(a - b) \cdot 4 = 20$, so $a - b = 5$.

4. Check: $a = \dfrac{4+5}{2} = \dfrac{9}{2}$, $b = \dfrac{4-5}{2} = -\dfrac{1}{2}$, and $a^2 - b^2 = \dfrac{81}{4} - \dfrac{1}{4} = 20$.

Answer: 5

16. HARD · NO CALCULATOR NEEDED

If $x > 0$ and $\sqrt{x} = x\sqrt{x} = x^2 = \sqrt{x^2}$, what is the value of x?

1. Take the first equality, $\sqrt{x} = x\sqrt{x}$. Since $x > 0$, divide both sides by $\sqrt{x}$: $1 = x$.

2. Check the whole chain at $x = 1$: $\sqrt{1} = 1$, $1 \cdot \sqrt{1} = 1$, $1^2 = 1$, $\sqrt{1^2} = 1$.

3. All four are equal, so $x = 1$.

Answer: 1

17. HARD · NO CALCULATOR NEEDED

If $s^2 = t^{-1}$ and $t^{1/3} = u^{-1/2}$, then what is the value of s in terms of u, for $s > 0$?

(A) $u^{3/4}$ ✓ (B) $u^{-3/4}$

(C) $u^{-1/2}$ (D) $u^{1/2}$

1. Cube both sides of $t^{1/3} = u^{-1/2}$: $t = u^{-1/2 \cdot 3} = u^{-\frac{3}{2}}$.

2. Then $s^2 = t^{-1} = \left(u^{-\frac{3}{2}}\right)^{-1} = u^{\frac{3}{2}}$.

3. Take the positive square root ($s > 0$): halve the exponent, $s = u^{\frac{3}{4}}$. Option (A).

Answer: (A)

18. HARD · NO CALCULATOR NEEDED

Let $\langle x \rangle$ be defined by $\langle x \rangle = \dfrac{3x+1}{x-2}$ for any x such that $x \neq 2$. Which of the following is equivalent to $\langle x \rangle - 3$?

(A) $\dfrac{6x-5}{x-2}$

(B) $\dfrac{7}{x-2} - 3$

(C) $\dfrac{3x-2}{x-2}$

(D) $\dfrac{7}{x-2}$ ✓

1. Write 3 with the same denominator: $3 = \dfrac{3(x-2)}{x-2}$.

2. Subtract the numerators and keep the brackets: $(3x+1) - 3(x-2) = 3x + 1 - 3x + 6 = 7$.

3. So $\langle x \rangle - 3 = \dfrac{7}{x-2}$.

4. Check at $x = 3$: $\dfrac{10}{1} - 3 = 7$ and $\dfrac{7}{1} = 7$. Option (D).

Answer: (D)

19. HARD · NO CALCULATOR NEEDED

Let a and b be numbers such that $a^4 = b^3$. Which of the following is equivalent to $a\sqrt{b}$?

(A) $b^{4/3}$

(B) $b^{3/2}$

(C) $b^{5/4}$ ✓

(D) $b^{7/6}$

1. From $a^4 = b^3$, take the fourth root: $a = b^{\frac{3}{4}}$.

2. Also $\sqrt{b} = b^{\frac{1}{2}}$.

3. Multiply, adding exponents: $a\sqrt{b} = b^{\frac{3}{4}+\frac{1}{2}} = b^{\frac{5}{4}}$. Option (C).

Answer: (C)

20. HARD · NO CALCULATOR NEEDED

What is the value of $\left(\dfrac{3x}{y}\right)\left(\dfrac{x^{-1}}{(3y)^{-1}}\right)$?

(A) 6

(B) 9 ✓

(C) 1

(D) 3

1. A negative exponent flips the base: $x^{-1} = \dfrac{1}{x}$ and $(3y)^{-1} = \dfrac{1}{3y}$.

2. Dividing by $\dfrac{1}{3y}$ means multiplying by $3y$: $\dfrac{x^{-1}}{(3y)^{-1}} = \dfrac{1}{x} \cdot 3y = \dfrac{3y}{x}$.

3. Multiply: $\dfrac{3x}{y} \cdot \dfrac{3y}{x} = \dfrac{9xy}{xy} = 9$. Option (B).

Answer: (B)

21. HARD · NO CALCULATOR NEEDED

Find the value of $\dfrac{\sqrt{c+7}}{c} \cdot c\sqrt{c}$ if $c = 9$.

1. The two c's cancel: $\dfrac{\sqrt{c+7}}{c} \cdot c\sqrt{c} = \sqrt{c+7} \cdot \sqrt{c}$.

2. Put in $c = 9$: $\sqrt{9+7} \cdot \sqrt{9} = \sqrt{16} \cdot 3$.

3. $4 \cdot 3 = 12$.

Answer: 12

22. HARD · CALCULATOR

Evaluate $\left(\dfrac{5}{a^2} + \dfrac{4}{3a}\right)^a$ for $a = -3$.

1. Put $a = -3$ inside: $\dfrac{5}{a^2} = \dfrac{5}{9}$ and $\dfrac{4}{3a} = \dfrac{4}{-9} = -\dfrac{4}{9}$.

2. The sum is $\dfrac{5}{9} - \dfrac{4}{9} = \dfrac{1}{9}$, so the expression is $\left(\dfrac{1}{9}\right)^{-3}$.

3. Enter all of it, with brackets round the negative exponent: type `(5/9+4/(3*(-3)))^(-3)` : it returns 729.

4. By hand: a negative exponent flips the fraction, so $\left(\dfrac{1}{9}\right)^{-3} = 9^3 = 729$.

Answer: 729

23. HARD · NO CALCULATOR NEEDED

For $x \neq -3$ and $x \neq 2$, the expression $\dfrac{x^2 - 2x - 15}{x^2 + x - 6}$ is equivalent to $\dfrac{x+a}{x+b}$, where a and b are constants. What is the value of $a - b$?

1. Factor the top: two numbers with product -15 and sum -2 are -5 and 3, so $x^2 - 2x - 15 = (x-5)(x+3)$.

2. Factor the bottom: product -6 and sum 1 give -2 and 3, so $x^2 + x - 6 = (x-2)(x+3)$.

3. Cancel $(x+3)$ (allowed since $x \neq -3$): $\dfrac{x-5}{x-2} = \dfrac{x+a}{x+b}$, so $a = -5$ and $b = -2$.

4. $a - b = -5 - (-2) = -3$.

Answer: -3

24. HARD · NO CALCULATOR NEEDED

If $a^{\frac{3}{4}} = 5$ and $a > 0$, what is the value of $a^{-\frac{3}{2}}$?

1. Look for a way to turn the exponent $\dfrac{3}{4}$ into $-\dfrac{3}{2}$: multiply by -2.

2. $a^{-3/2} = \left(a^{3/4}\right)^{-2}$ because $\dfrac{3}{4} \cdot (-2) = -\dfrac{3}{2}$.

3. So $a^{-3/2} = 5^{-2} = \dfrac{1}{25} = \dfrac{1}{25}$, which is 0.04 as a decimal.

Answer: $\dfrac{1}{25}$ **(or** 0.04**)**

25. HARD · NO CALCULATOR NEEDED

If $a^2 + b^2 = 4ab$, where $a \neq b$ and $b \neq 0$, what is the value of $\dfrac{a+b}{a-b}$?

(A) 2 (B) 3

(C) 6 **(D) $\pm\sqrt{3}$ ✓**

1. Add and subtract $2ab$: $(a+b)^2 = a^2 + b^2 + 2ab = 4ab + 2ab = 6ab$ and $(a-b)^2 = a^2 + b^2 - 2ab = 4ab - 2ab = 2ab$.

2. Divide: $\left(\dfrac{a+b}{a-b}\right)^2 = \dfrac{6ab}{2ab} = 3$ ($ab \neq 0$ because $b \neq 0$ and $a = 0$ would force $b = 0$).

3. So $\dfrac{a+b}{a-b} = \pm\sqrt{3}$. Both signs happen: with $b = 1$, $a = 2 \pm \sqrt{3}$ gives 1.7321 and -1.7321.

4. The only option that lists both is (D).

Answer: (D)

26. HARD · NO CALCULATOR NEEDED

The equation $\dfrac{ax+b}{x+2} = 3 - \dfrac{11}{x+2}$ is true for all values of x not equal to -2, where a and b are constants. What is the value of ab?

1. Put the right side over the denominator $x + 2$: $3 - \dfrac{11}{x+2} = \dfrac{3(x+2) - 11}{x+2}$.

2. The numerator is $3x + 6 - 11 = 3x - 5$.

3. Match $ax + b$ with $3x - 5$: $a = 3$ and $b = -5$.

4. $ab = 3 \cdot (-5) = -15$.

Answer: -15

27. HARD · DESMOS

The expression $10x^4 + 33x^2 + 20$ can be rewritten as $(5x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(5x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + d$?

1. Expand $(5x^2 + c)(2x^2 + d) = 10x^4 + (5d + 2c)x^2 + cd$. Matching gives $cd = 20$ and $5d + 2c = 33$ (the same for a, b).

2. Put $d = \dfrac{20}{c}$ into $5d + 2c = 33$ and multiply by c: $100 + 2c^2 = 33c$, so $2c^2 - 33c + 100 = 0$. Type `y=2x^2-33x+100` (letting x stand for c).

3. Tap the zeros: (4, 0) and (12.5, 0). The whole-number zero 4 is a (with $b = 5$). The other zero is $c = 12.5$, and then type `20/12.5`: it returns 1.6, so $d = 1.6$ is a positive nonintegral value.

4. Type `4+1.6`: it returns 5.6, which is $\dfrac{28}{5}$.

Answer: $\dfrac{28}{5}$ **(or** 5.6**)**

28. HARD · DESMOS

$$\frac{40x+100}{2x^2-5x-25}+\frac{x^2y-5xy}{4xy}$$

Which of the following is equivalent to the expression above for $x>5$, $y>0$?

(A) $\dfrac{x^2-10x+105}{4(x-5)}$ ✓

(B) $\dfrac{x^2-10x+105}{4(2x+5)}$

(C) 5

(D) $\dfrac{x-1}{x+15}$

1. The letter y cancels in the second fraction (since $y>0$), so test with $y=2$. Type `f(x)=(40x+100)/(2x^2-5x-25)+(2x^2-10x)/(8x)` and look at $x>5$.
2. Type each option as its own graph, for example `g(x)=(x^2-10x+105)/(4(x-5))` and `h(x)=(x^2-10x+105)/(4(2x+5))`. Graph g lies exactly on f; graph h does not.
3. Confirm at $x=7$: type `f(7)`: it returns 10.5; type `g(7)`: it returns 10.5 (same); type `h(7)`: it returns 1.1053 (different). Option (A).
4. Algebra: $\dfrac{40x+100}{2x^2-5x-25}=\dfrac{20(2x+5)}{(2x+5)(x-5)}=\dfrac{20}{x-5}$ and the second fraction is $\dfrac{x-5}{4}$. Over $4(x-5)$ the numerator is $80+(x-5)^2=x^2-10x+105$.

Answer: (A)

29. HARD · NO CALCULATOR NEEDED

For real numbers p and q with $p^2+4q^2=72$, the product $(2x+p)(x+q)$ is equivalent to $2x^2+bx+c$, where b and c are constants. What is the greatest possible value of c?

1. Expand: $(2x+p)(x+q)=2x^2+(2q+p)x+pq$, so $c=pq$.
2. Let $u=2q$, so $p^2+u^2=72$ and $c=pq=\dfrac{pu}{2}$. Since $(p-u)^2\geq 0$, we have $2pu\leq p^2+u^2=72$, so $pu\leq 36$.
3. Therefore $c=\dfrac{pu}{2}\leq\dfrac{36}{2}=18$, with equality when $p=u$.
4. That happens at $p=6$ and $q=3$: $p^2+4q^2=36+36=72$ and $pq=18$.

Answer: 18

30. HARD · NO CALCULATOR NEEDED

For constants p, q and c, the equation

$$(2x-3)(x^2+px+q)=2x^3-5x^2+cx+9$$

is true for all real values of x. What is the value of c?

1. Expand the left side: $(2x-3)(x^2+px+q)=2x^3+(2p-3)x^2+(2q-3p)x-3q$.
2. Match the x^2 terms: $2p-3=-5$, so $p=-1$. Match the constant terms: $-3q=9$, so $q=-3$.
3. Match the x terms: $c=2q-3p=2(-3)-3(-1)=-6+3=-3$.

Answer: -3

31. HARD · NO CALCULATOR NEEDED

For a positive real number x, $\dfrac{\sqrt[3]{x}\cdot\sqrt[4]{x^3}}{\sqrt[6]{x}}=\sqrt[3]{k}$, where $k=2^{11}$. Which of the following is equal to x?

(A) $2^{\frac{44}{3}}$ **(B) 2^4 ✓**

(C) $2^{\frac{11}{3}}$ (D) $2^{\frac{121}{36}}$

1. Combine the exponents of x: $\dfrac{1}{3}+\dfrac{3}{4}-\dfrac{1}{6}=\dfrac{4+9-2}{12}=\dfrac{11}{12}$.

2. So the left side is $x^{\frac{11}{12}}$, and the right side is $\sqrt[3]{k}=\left(2^{11}\right)^{1/3}=2^{\frac{11}{3}}$.

3. Raise both sides to the power $\dfrac{12}{11}$: $x=2^{\frac{11}{3}\cdot\frac{12}{11}}=2^4$. Option (B).

Answer: (B)

32. HARD · NO CALCULATOR NEEDED

For $x>0$ and any positive integer n, which of the following is equivalent to $\dfrac{\left(x^{2n+1}\right)^3+4x^{6n+3}-2x\left(x^{3n+1}\right)^2}{x^{6n+1}}$?

(A) $3x^4$ (B) $3x^{12n+4}$

(C) $3x^2$ ✓ (D) $5x^2$

1. Simplify each power: $\left(x^{2n+1}\right)^3=x^{6n+3}$, and $2x\left(x^{3n+1}\right)^2=2x\cdot x^{6n+2}=2x^{6n+3}$.

2. The numerator is $x^{6n+3}+4x^{6n+3}-2x^{6n+3}=(1+4-2)x^{6n+3}=3x^{6n+3}$.

3. Divide by x^{6n+1}: $3x^{(6n+3)-(6n+1)}=3x^2$.

4. Check with $n=1$, $x=2$: the fraction is $\dfrac{512+2048-1024}{128}=12$ and $3x^2=12$. Option (C).

Answer: (C)

33. HARD · NO CALCULATOR NEEDED

For real numbers $k\neq 3$ and $x\neq 0$:

$$\frac{4(k-3)}{x(k-3)}+\frac{2}{x}-\frac{3}{4}=\frac{3}{8}$$

What is the value of x?

1. Since $k\neq 3$, the factor $(k-3)$ cancels: $\dfrac{4(k-3)}{x(k-3)}=\dfrac{4}{x}$.

2. Add the terms with x: $\dfrac{4}{x}+\dfrac{2}{x}=\dfrac{6}{x}$, so $\dfrac{6}{x}-\dfrac{3}{4}=\dfrac{3}{8}$.

3. $\dfrac{6}{x}=\dfrac{3}{8}+\dfrac{3}{4}=\dfrac{9}{8}$.

4. Cross-multiply: $x=\dfrac{6\cdot 8}{9}=\dfrac{16}{3}$.

Answer: $\dfrac{16}{3}$

34. HARD · CALCULATOR

The function f is defined by $f(x) = \dfrac{9^x \cdot 12^{x+1}}{8^x}$. Which of the following expressions is equivalent to $f(x)$ and displays the value of $f(2)$ as a coefficient?

(A) $2187\left(\dfrac{27}{2}\right)^{x-2}$ ✓

(B) $2187\left(\dfrac{27}{2}\right)^{x+2}$

(C) $162\left(\dfrac{27}{2}\right)^{x-1}$

(D) $\dfrac{729}{4}\left(\dfrac{27}{2}\right)^{x-2}$

1. Combine the powers: $\dfrac{9^x \cdot 12^{x+1}}{8^x} = 12\left(\dfrac{9 \cdot 12}{8}\right)^x$. Type `9*12/8`: it returns 13.5, which is $\dfrac{27}{2}$. So $f(x) = 12\left(\dfrac{27}{2}\right)^x$.

2. Find $f(2)$: type `12*13.5^2`: it returns 2187.

3. To show $f(2)$ as the coefficient, write $f(x) = f(2)\left(\dfrac{27}{2}\right)^{x-2}$, that is $2187\left(\dfrac{27}{2}\right)^{x-2}$.

4. Why not the others: type `13.5^2`: it returns 182.25 is only $\left(\dfrac{27}{2}\right)^2$ (the factor 12 is missing), and type `12*13.5`: it returns 162 is $f(1)$, not $f(2)$. Option (A).

Answer: (A)

35. HARD · NO CALCULATOR NEEDED

For integers a, b, c, and d, $(ax + b)(cx + d) = 6x^2 + kx + 5$ for all values of x, where k is a constant. Which of the following must be true?

I. k is odd
II. When k is divided by 4, the remainder is 1
III. k is not divisible by 5

(A) II and III only

(B) I, II, and III

(C) I only

(D) I and III only ✓

1. Matching the constants: $bd = 5$, so b, d are odd (one of $\pm 1, \pm 5$). Matching the x^2 terms: $ac = 6$. The middle coefficient is $k = ad + bc$.

2. Try $b = 1$, $d = 5$ and the pairs $(a, c) = (1, 6), (2, 3), (3, 2), (6, 1)$: $k = 5a + c$ gives $11, 13, 17, 31$. The other choices of b, d give these values or their negatives.

3. I: exactly one of a, c is even and b, d are odd, so $k = ad + bc$ is always odd (true). II: 11 divided by 4 leaves remainder 3, not 1 (false). III: none of $11, 13, 17, 31$ is divisible by 5 (true).

4. Statements I and III only. Option (D).

Answer: (D)

PART B · SELF-ASSESSMENT — 22 PROBLEMS

36. Medium · No calculator needed

Which of the following is equivalent to $\dfrac{x^2-9}{x-3}$, for $x \neq 3$?

(A) x^2-3 (B) $x+9$

(C) $x+3$ ✓ (D) $x-3$

1. Factor the top as a difference of squares: $x^2-9=(x-3)(x+3)$.
2. Cancel $(x-3)$, which is allowed because $x \neq 3$: the result is $x+3$.
3. Check at $x=5$: $\dfrac{25-9}{5-3}=8$ and $5+3=8$. Option (C).

Answer: (C)

37. Medium · No calculator needed

Which of the following is equivalent to $\dfrac{x^2+x-12}{x+4}$, for $x \neq -4$?

(A) x^2-3 **(B) $x-3$ ✓**

(C) $x+3$ (D) $x-4$

1. Factor the top: two numbers with product -12 and sum 1 are -3 and 4, so $x^2+x-12=(x+4)(x-3)$.
2. Cancel $(x+4)$, which is allowed because $x \neq -4$: the result is $x-3$.
3. Check at $x=2$: $\dfrac{4+2-12}{2+4}=-1$ and $2-3=-1$. Option (B).

Answer: (B)

38. Medium · No calculator needed

If a and b are positive numbers such that $a^5=b^2$, which of the following is equivalent to $b\sqrt{a}$?

(A) $b^{9/4}$ (B) $b^{1/5}$

(C) $b^{9/5}$ **(D) $b^{6/5}$ ✓**

1. From $a^5=b^2$, take the fifth root: $a=b^{\frac{2}{5}}$.
2. Then $\sqrt{a}=\left(b^{\frac{2}{5}}\right)^{1/2}=b^{\frac{1}{5}}$.
3. Multiply by $b=b^1$ and add exponents: $b\sqrt{a}=b^{1+\frac{1}{5}}=b^{\frac{6}{5}}$. Option (D).

Answer: (D)

39. Medium · No calculator needed

What is the quotient when $2x^3 - 3x^2 - 11x + 6$ is divided by $x - 3$ (the division has no remainder)?

(A) $2x^2 + 3x - 2$ ✓ (B) $2x^2 - 3x - 2$

(C) $2x^2 + 3x + 2$ (D) $2x^2 - 9x + 16$

1. Use synthetic division with the root 3 and the coefficients 2, -3, -11, 6.
2. Bring down 2. Then multiply by 3 and add to the next coefficient each time: $-3 + 6 = 3$, then $-11 + 9 = -2$, then $6 + (-6) = 0$.
3. The remainder is 0, as the question says, and the quotient is $2x^2 + 3x - 2$.
4. Check: $(x - 3)(2x^2 + 3x - 2) = 2x^3 - 3x^2 - 11x + 6$. Option (A).

Answer: (A)

40. Medium · No calculator needed

Which expression is equivalent to $4\,(2x + 7) - 5x$?

(A) $13x + 28$ **(B)** $3x + 28$ ✓

(C) $3x + 2$ (D) $13x + 7$

1. Distribute the 4: $4(2x + 7) = 8x + 28$.
2. Subtract $5x$: $8x + 28 - 5x = 3x + 28$. Option (B).
3. Check at $x = 1$: $4(9) - 5 = 31$ and $3 + 28 = 31$.

Answer: (B)

41. Medium · No calculator needed

Which expression is equivalent to $\left(5x^2 + 3x - 9\right) - \left(2x^2 - 7x + 4\right)$?

(A) $7x^2 - 4x - 5$ (B) $7x^2 + 10x - 5$

(C) $3x^2 + 10x - 13$ ✓ (D) $3x^2 - 4x - 13$

1. Subtracting a bracket changes every sign inside it: $5x^2 + 3x - 9 - 2x^2 + 7x - 4$.
2. Combine like terms: x^2: $5 - 2 = 3$. x: $3 + 7 = 10$. Constants: $-9 - 4 = -13$.
3. The result is $3x^2 + 10x - 13$. Option (C).
4. Check at $x = 1$: $(5 + 3 - 9) - (2 - 7 + 4) = 0$ and $3 + 10 - 13 = 0$.

Answer: (C)

42. Medium · No calculator needed

An urban planner models the area of a rectangular section of a park as $3x^2 + ax$ square meters, where a is a constant. This expression is equivalent to $x\,(3x + 11)$. What is the value of a?

(A) 11 ✓ (B) 14

(C) 3 (D) 8

1. Expand $x(3x + 11) = 3x^2 + 11x$.
2. Match with $3x^2 + ax$: the x-coefficients must be equal, so $a = 11$.
3. Check at $x = 1$: $3(1)^2 + 11(1) = 14$ and $1 \cdot (3 + 11) = 14$. Option (A).
Answer: (A)

43. Medium · No calculator needed

The sum of $-3x^2 + 5x + 18$ and $6x^2 - 2x - 9$ can be written in the form $ax^2 + bx + c$, where a, b, and c are constants. What is the value of $a + b + c$?
1. Add like terms: x^2: $-3 + 6 = 3$. x: $5 + (-2) = 3$. Constants: $18 + (-9) = 9$.
2. So $a = 3$, $b = 3$, $c = 9$ and $a + b + c = 3 + 3 + 9 = 15$.
3. Shortcut: $a + b + c$ is the value of the sum at $x = 1$: $(-3 + 5 + 18) + (6 - 2 - 9) = 20 + (-5) = 15$.
Answer: 15

44. Medium · No calculator needed

Which of the following is equivalent to $\dfrac{2x^2 + 8x}{2x}$, for $x \neq 0$?

(A) $x + 8$ (B) $2x + 4$

(C) $x^2 + 4$ **(D)** $x + 4$ ✓

1. Factor the top: $2x^2 + 8x = 2x(x + 4)$.
2. Cancel $2x$ (allowed since $x \neq 0$): the result is $x + 4$.
3. Check at $x = 1$: $\dfrac{2 + 8}{2} = 5$ and $1 + 4 = 5$. Option (D).
Answer: (D)

45. Medium · No calculator needed

If $x > 0$ and $\sqrt{x\sqrt{x}} = x^a$, what is the value of a?
1. Inside the root: $x\sqrt{x} = x^1 \cdot x^{1/2} = x^{\frac{3}{2}}$.
2. Take the square root (halve the exponent): $\sqrt{x^{\frac{3}{2}}} = x^{\frac{3}{4}}$.
3. So $a = \dfrac{3}{4}$ (or 0.75 as a decimal).
Answer: $\dfrac{3}{4}$

46. Medium · No calculator needed

If $a^{b/5} = 32$ for positive integers a and b, where a is a prime number, what is the value of b?

1. Write 32 as a power of a prime: $32 = 2^5$.

2. The base a is prime, and $a^{b/5} = 2^5$ only works for $a = 2$ (a different prime would give a different prime factor).

3. Then $\frac{b}{5} = 5$, so $b = 25$.

4. Check: $2^{25/5} = 2^5 = 32$.

Answer: 25

47. Medium · No calculator needed

Which of the following is a factor of the polynomial $\left(x^2 - 9\right)(x + 7)$?

(A) $x + 5$ (B) $x - 2$

(C) $x + 3$ ✓ (D) $x - 4$

1. Factor the difference of squares: $(x^2 - 9)(x + 7) = (x - 3)(x + 3)(x + 7)$.

2. So the factors are $x - 3$, $x + 3$ and $x + 7$. Only $x + 3$ is in the list.

3. Check the others by putting in their zeros: at $x = 2$ the polynomial is -45, at $x = 4$ it is 77, at $x = -5$ it is 32 (none is 0), while at $x = -3$ it is 0. Option (C).

Answer: (C)

48. Medium · No calculator needed

If $k = m^2n^2$, k is not 0, and the values of m and n are both tripled, then the value of k increases by which of the following factors?

(A) 27 **(B) 81 ✓**

(C) 3 (D) 9

1. Triple both: $k_{\text{new}} = (3m)^2(3n)^2$.

2. Square each: $(3m)^2 = 9m^2$ and $(3n)^2 = 9n^2$.

3. So $k_{\text{new}} = 9 \cdot 9\, m^2n^2 = 81k$. The factor is 81. Option (B).

Answer: (B)

49. Medium · No calculator needed

In the equation $\sqrt{200} + \sqrt{200} = m\sqrt{n}$, m and n are integers such that $m > n$ and n is not divisible by the square of any integer greater than 1. What is the value of m?

1. Simplify one root: $\sqrt{200} = \sqrt{100 \cdot 2} = 10\sqrt{2}$ (100 is the largest perfect square dividing 200).

2. Add the two equal roots: $10\sqrt{2} + 10\sqrt{2} = 20\sqrt{2}$.

3. So $m = 20$ and $n = 2$: n has no square factor and $m > n$.

4. Check: $20^2 \cdot 2 = 800$ and $(2\sqrt{200})^2 = 4 \cdot 200 = 800$.

Answer: 20

50. Medium · No calculator needed

The value of y decreased by 5 is directly proportional to the value of x increased by 3. If $y = 7$ when $x = 1$, what is the value of x when $y = 19$?

1. Write the statement as an equation: $y - 5 = k(x + 3)$.

2. Use $y = 7$ when $x = 1$: $7 - 5 = k(1 + 3)$, so $2 = 4k$ and $k = \frac{1}{2}$.

3. Now $y = 19$: $19 - 5 = \frac{1}{2}(x + 3)$, so $14 = \frac{1}{2}(x + 3)$ and $x + 3 = 28$.

4. $x = 28 - 3 = 25$.

Answer: 25

51. Medium · Desmos

The expression $\dfrac{11p^2 + 11q^2}{9p^2 + 30p + 9}$ is undefined when

(A) $p = 0$ or $q = 0$

(B) $q = -\frac{1}{3}$ or $q = -3$

(C) $p = -3$ or $q = -\frac{1}{3}$

(D) $p = -\frac{1}{3}$ or $p = -3$ ✓

1. A fraction is undefined when its denominator is 0. The numerator does not matter, and the denominator has only p in it, so q cannot make it undefined.

2. Type `y=9x^2+30x+9` (letting x stand for p) and tap the zeros.

3. The zeros are $(-3,\ 0)$ and $(-0.3333,\ 0)$, so $p = -3$ or $p = -\frac{1}{3}$.

4. Check by factoring: $9p^2 + 30p + 9 = 3(3p + 1)(p + 3)$, and $3(3p + 1)(p + 3) = 9x^2 + 30x + 9$. Option (D).

Answer: (D)

52. Medium · No calculator needed

For $x > 0$, which expression is equivalent to $\sqrt{(9x^2)^3 \cdot x^6}$?

(A) $27x^6$ ✓

(B) $27x^9$

(C) $81x^6$

(D) $3x^6$

1. Cube the first bracket: $(9x^2)^3 = 9^3x^6 = @a@x^6$.

2. Multiply by x^6: $729x^6 \cdot x^6 = 729x^{12}$.

3. Take the square root ($x > 0$): $\sqrt{729} = 27$ and $\sqrt{x^{12}} = x^6$. Result: $27x^6$. Option (A).

Answer: (A)

53. Medium · No calculator needed

If $\left(\sqrt{p}\right)^6 = q^{-3}$ and $q = \frac{1}{4}$, what is the value of p?

1. Simplify the left side: $\left(\sqrt{p}\right)^6 = \left(p^{1/2}\right)^6 = p^3$.

2. Simplify the right side with $q = \frac{1}{4}$: $q^{-3} = \left(\frac{1}{4}\right)^{-3} = 4^3 = 64$.

3. So $p^3 = 64$ and $p = 4$. Check: $\left(\sqrt{4}\right)^6 = 2^6 = 64$.

Answer: 4

54. MEDIUM · NO CALCULATOR NEEDED

$\dfrac{\sqrt[3]{x^7}}{\sqrt{x}} = x^{a/b}$ for $x > 1$, where a and b are positive integers with no common factor. What is the value of $\dfrac{a}{b}$?

1. Write both as powers: $\sqrt[3]{x^7} = x^{\frac{7}{3}}$ and $\sqrt{x} = x^{\frac{1}{2}}$.

2. Dividing subtracts exponents: $\dfrac{7}{3} - \dfrac{1}{2} = \dfrac{14-3}{6} = \dfrac{11}{6}$.

3. So $\dfrac{a}{b} = \dfrac{11}{6}$, and 11 and 6 have no common factor.

Answer: $\dfrac{11}{6}$

55. MEDIUM · NO CALCULATOR NEEDED

If $x + \dfrac{1}{x} = 5$, what is the value of $x^2 + \dfrac{1}{x^2}$?

1. Square both sides of $x + \dfrac{1}{x} = 5$.

2. $\left(x + \dfrac{1}{x}\right)^2 = x^2 + 2 \cdot x \cdot \dfrac{1}{x} + \dfrac{1}{x^2} = x^2 + 2 + \dfrac{1}{x^2}$, and the left side is $5^2 = 25$.

3. So $x^2 + \dfrac{1}{x^2} = 25 - 2 = 23$.

Answer: 23

56. MEDIUM · NO CALCULATOR NEEDED

$$\frac{2}{x-2} + \frac{3}{x+5} = \frac{rx+t}{(x-2)(x+5)}$$

The equation above is true for all $x \neq 2, -5$, where r and t are constants. What is the value of rt?

(A) 15 **(B) 20 ✓**

(C) 60 (D) -20

1. Use the common denominator $(x-2)(x+5)$: $\dfrac{2(x+5)+3(x-2)}{(x-2)(x+5)}$.

2. The numerator is $2x + 10 + 3x - 6 = 5x + 4$.

3. Match with $rx + t$: $r = 5$ and $t = 4$.

4. $rt = 5 \cdot 4 = 20$. Option (B).

Answer: (B)

57. MEDIUM · NO CALCULATOR NEEDED

$\dfrac{\left(\sqrt[3]{x^2}\right)^5}{x^2} = x^{a/b}$ for $x > 1$, where a and b are positive integers with no common factor. What is the value of $\dfrac{a}{b}$?

1. Write the root as a power: $\sqrt[3]{x^2} = x^{\frac{2}{3}}$.

2. Raise to the fifth power, multiplying exponents: $\left(x^{\frac{2}{3}}\right)^5 = x^{\frac{10}{3}}$.

3. Divide by x^2, subtracting exponents: $\dfrac{10}{3} - 2 = \dfrac{10-6}{3} = \dfrac{4}{3}$. So $\dfrac{a}{b} = \dfrac{4}{3}$, with no common factor.

Answer: $\dfrac{4}{3}$